# MDSimulationAgent Short Usage Guide

Quick start, key commands, outputs, limitations, and improvement ideas

**Short version:** install **mdagent**, install GROMACS, run a config or ask Claude Code to run one of the bundled skills, then inspect **REPORT.md** plus the per-step JSON artifacts.

## What It Does

MDSimulationAgent is a Python CLI and Claude Code skill bundle for agent-driven GROMACS workflows. It takes a PDB ID or local structure, prepares a soluble protein system, builds topology, solvates and neutralizes it, runs EM/NVT/NPT/production, performs basic analysis, and writes a readiness report.

## Quick Start

```
brew install uv
uv tool install --force git+https://github.com/<user>/MDSimulationAgent@v0.1.0
brew install gromacs
mdagent install-skills --user

mdagent run-workflow --runs-root ./runs --pdb-id 1AKI --run-id demo
mdagent inspect --run-root ./runs/demo
```

## Starter Kit Option

```
mdagent init-project ./my-md-project
cd ./my-md-project
./verify.sh
./verify.sh --run-smoke
```

## Core Commands

| Command | Use |
| --- | --- |
| mdagent doctor --gmx-required | Check the environment before running MD. |
| mdagent prep-structure | Ingest, classify, and prep only; no GROMACS required. |
| mdagent run-workflow | Run the full or partial pipeline. |
| mdagent inspect | Print index.json status and REPORT.md. |
| mdagent visualize | Generate viewer scripts and optional static checkpoint PNGs. |
| mdagent init-project | Create a runnable starter directory. |
| mdagent pack-bundle | Create a handoff bundle for another machine. |

## Useful Flags

```
--config path/to/run_config.json
--runs-root ./runs
--run-id meaningful-name
--pdb-id 1AKI
--structure-path /path/to/protein.pdb
--stop-after prep|topology|solvation|em|nvt|npt
--pipeline-mode tutorial_reproduction|general_md_prep
```

```
--viz-mode disabled|default|requested
```

## What To Inspect

| File | Why it matters |
|---|---|
| REPORT.md | Readiness headline: ready, ready_with_warnings, blocked, or not_validated. |
| index.json | Step statuses, artifacts, attempts, and fingerprints. |
| step_report.json | Inputs, outputs, warnings, executor calls, and failure reason per step. |
| classification.json | Supported/unsupported chemistry decision. |
| charge_accounting.json | Ion insertion and final charge validation. |
| em_convergence.json | Whether minimization converged, needs longer, diverged, or stuck. |
| analysis.json | RMSD, Rg, RMSF, H-bonds, temperature, pressure, and density summaries. |

## Limits To Remember

- Best-supported systems are soluble protein-only. Ligands, nucleic acids, membranes, glycans, cofactors, and many modifications are outside the current supported path.
- GROMACS prompt automation is version-sensitive; current recognizers target modern 2026.x prompt shapes.
- PROPKA is optional and conditional. Without the extra package, the workflow falls back to fixed pH-7 defaults.
- The starter smoke run and default tutorial-scale runs validate the workflow, not scientific convergence.
- Execution is local; HPC/cloud execution is not wired yet.
- Analysis and visualization are sanity checks, not full downstream science.

## Best Next Improvements

- Add ligand/cofactor/membrane/nucleic-acid support and richer structure repair.
- Build versioned GROMACS prompt catalogs and broader force-field tests.
- Add remote executor support for Slurm/cloud/GPU runs.
- Improve config UX, force-field/water compatibility checks, plots, and richer analysis reports.
- Expand golden-path tests beyond lysozyme and document release-ready install URLs.